Ejercicios Certamen 2 Mecánica Clásica Avanzada

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1 Problema 1

A point-like nonrelativistic particle of mass m scatters in the time-dependent spherically symmetric harmonic potential

$$U(r,t) = \frac{k(t)}{2} r^2, \qquad k(t) = \frac{\kappa}{t^2}, \qquad \kappa > 0 \tag{1.1}$$

(a) Identify all continuous symmetries of the action. Analyze if there are symmetries of type

$$\mathbf{r} \to \alpha \mathbf{r}, \qquad t \to \beta t, \qquad \alpha, \beta = \text{const.} \tag{1.2}$$

which do not change the equations of motion.

El lagrangiano será

$$L = \frac{1}{2} m \left(\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \dot{\phi}^2 \sin^2\theta \right) + \frac{\kappa}{2t^2} r^2 \tag{1.3}$$

La variable ϕ es cíclica, por lo que M_z será conservado. Aun más, el sistema es esféricamente simétrico, por lo que todo el vector de momento angular se conservará:

$$\mathbf{M} = \text{const.} \tag{1.4}$$

Reemplazando $\mathbf{r} \to \alpha \mathbf{r}, t \to \beta t$ en las derivadas,

$$\frac{\partial r}{\partial t} \to \frac{\partial \alpha r}{\partial \beta t} = \frac{\alpha}{\beta} \frac{\partial r}{\partial t} \tag{1.5}$$

Luego en el lagrangiano

$$\begin{aligned}
L &= \frac{1}{2} m \left(\frac{\alpha^2}{\beta^2} \dot{r}^2 + \frac{\alpha^2}{\beta^2} r^2 \dot{\theta}^2 + \frac{\alpha^2}{\beta^2} r^2 \dot{\phi}^2 \sin^2\theta \right) + \frac{\kappa}{2\beta^2 t^2} \alpha^2 r^2 \\
L &= \frac{\alpha^2}{\beta^2} \left[\frac{1}{2} m \left(\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \dot{\phi}^2 \sin^2\theta \right) + \frac{\kappa}{2t^2} r^2 \right] \\
L &= \frac{\alpha^2}{\beta^2} L
\end{aligned}$$

Ya que multiplicar un lagrangiano por una cantidad constante no cambia las ecuaciones de movimiento, tendremos una simetría sí y solo sí $\alpha, \beta \in \mathbb{R}, \beta \neq 0$.

(b) Construct the integrals of motion which might exist according to Noether's theorem. Demonstrate that these quantities are conserved using the equations of motion.

El momento angular vendrá dado por

$$\mathbf{M} = m\mathbf{r} \times \dot{\mathbf{r}} = \text{const.} \tag{1.6}$$

Para la simetría de escala, escribiendo las transformaciones en notación con ε